Physics P3: Gravity, Orbits, and Waves

MAIS Physics & Astronomy Club (PAC)

Newton’s discovery that the same inverse-square law governs a falling apple and the orbiting Moon turned astronomy into a branch of physics. Together with wave physics and the Doppler shift, orbital mechanics lets us weigh distant planets, stars, and even invisible companions across light-years of empty space.

1. Universal Gravitation and Circular Orbits

Two point masses $\mathbf{M}$ and $\mathbf{m}$ separated by distance $\mathbf{r}$ attract with force $\mathbf{F} = \mathbf{GMm}/\mathbf{r}^2$, with potential energy $\mathbf{U}(\mathbf{r}) = -\mathbf{GMm}/\mathbf{r}$ (choosing $\mathbf{U}(\infty) = 0$). By Newton’s shell theorem, any spherically symmetric body acts on external objects as if all of its mass were concentrated at its center. For a circular orbit of radius $\mathbf{r}$ around a much heavier mass $\mathbf{M} \gg \mathbf{m}$, equating gravitational pull to centripetal force gives

$$(\mathbf{GMm})/\,(\mathbf{r}^2) = (\mathbf{mv}^2)/\,(\mathbf{r}) \Rightarrow \mathbf{v} = \sqrt{((\mathbf{GM})/\,(\mathbf{r}))},\ \mathbf{P}^2 = (4\pi^2)/\,(\mathbf{GM})\ \mathbf{r}^3. \tag{1}$$

[1] Problem 1 (Geostationary satellites). A weather or communications satellite in geostationary orbit stays fixed above a single longitude on Earth’s equator ($\mathbf{G} = 6.67 \times 10^{-11}$ N·m^2/kg^2, $\mathbf{M}_\oplus = 5.97 \times 10^{24}$ kg, $\mathbf{R}_\oplus = 6.37 \times 10^{6}$ m).

(a) Explain why the satellite’s orbital period must equal Earth’s **sidereal** rotation period ($\mathbf{P} = 86{,}164$ s) rather than the 86,400 s solar day, and explain why a stationary orbit directly above Ulaanbaatar (latitude 47.9° N) is impossible under gravity alone.

(b) Compute the satellite’s orbital radius $\mathbf{r}$ and its altitude $\mathbf{h} = \mathbf{r} - \mathbf{R}_\oplus$ above Earth’s surface (in km).

2. Elliptical Orbits and Orbital Energy

Bound orbits under an inverse-square force are ellipses with the central mass $\mathbf{M}$ at one focus. If the ellipse has semi-major axis $\mathbf{a}$ and eccentricity $\mathbf{e}$ ($0 \le \mathbf{e} < 1$), its closest approach (periapsis) is $\mathbf{r_p} = \mathbf{a}(1 - \mathbf{e})$ and its farthest point (apoapsis) is $\mathbf{r_a} = \mathbf{a}(1 + \mathbf{e})$. Remarkably, both total mechanical energy $\mathbf{E}$ and orbital period $\mathbf{P}$ depend **only** on $\mathbf{a}$, not on eccentricity:

$$\mathbf{E} = \mathbf{K} + \mathbf{U} = -(\mathbf{GMm})/\,(2\mathbf{a}),\ \mathbf{v}^2 = \mathbf{GM}\,((2)/\,(\mathbf{r}) - (1)/\,(\mathbf{a})). \tag{2}$$

Remark (The vis-viva equation). The second relation in (2) follows immediately from $\mathbf{E} = (1)/\,(2)\mathbf{mv}^2 - \mathbf{GMm}/\mathbf{r} = -\mathbf{GMm}/(2\mathbf{a})$. Check two limits: for a circle ($\mathbf{r} = \mathbf{a}$) it gives $\mathbf{v}^2 = \mathbf{GM}/\mathbf{r}$; for an unbound parabolic escape trajectory ($\mathbf{a} \to \infty$) it gives escape speed $\mathbf{v}_{\text{esc}} = \sqrt{(2\mathbf{GM}/\mathbf{r})} = \sqrt{2}\ \mathbf{v}_{\text{circ}}$.

[2] Problem 2 (The atmospheric drag paradox). A low-Earth satellite in a nearly circular orbit of radius $\mathbf{r}$ experiences a very weak atmospheric drag force opposing its motion, losing a small amount of mechanical energy per orbit while its path remains nearly circular.

(a) Does the satellite's orbital speed $\mathbf{v}$ slow down or **speed up** as drag removes mechanical energy? Prove your answer using equation (2).

(b) Resolve the apparent paradox using the work–energy theorem: if drag opposes the velocity vector, what force does positive work larger than the negative work of drag?

[2] Problem 3 (Journey to Mars). Model Earth and Mars as moving in coplanar circular orbits around the Sun of radii $\mathbf{r}_1 = 1.00$ AU and $\mathbf{r}_2 = 1.52$ AU. A fuel-efficient Hohmann transfer places a spacecraft onto an elliptical orbit tangent to Earth's orbit at perihelion ($\mathbf{r_p} = \mathbf{r}_1$) and tangent to Mars's orbit at aphelion ($\mathbf{r_a} = \mathbf{r}_2$).

(a) Find the semi-major axis $\mathbf{a}$ of the transfer ellipse (in AU) and use Kepler's third law ($\mathbf{P}^2 = \mathbf{a}^3$ in years and AU) to calculate the one-way travel time to Mars in days.

(b) Given Earth's orbital speed $\mathbf{v}_\oplus = 29.8$ km/s, use the vis-viva equation to find the speed boost $\Delta\mathbf{v}_1$ needed to enter the transfer ellipse from Earth's orbit (ignoring Earth's local gravity).

3. Waves and the Doppler Effect

Light in vacuum travels at $\mathbf{c} = \lambda\nu = 3.00 \times 10^8$ m/s. When a source moves with radial velocity $\mathbf{v_r} \ll \mathbf{c}$ along the line of sight (taking $\mathbf{v_r} > 0$ for recession and $\mathbf{v_r} < 0$ for approach), every spectral line of rest wavelength λ_0 shifts by

$$(\Delta\lambda)/(\lambda_0) = (\lambda - \lambda_0)/(\lambda_0) \approx (\mathbf{v_r})/(\mathbf{c}). \tag{3}$$

[3] Problem 4 (Detecting unseen worlds via Doppler wobble). A star of mass $\mathbf{M}_\star$ and a planet of mass $\mathbf{m_p} \ll \mathbf{M}_\star$ orbit their common center of mass in circular orbits of period $\mathbf{P}$, viewed edge-on from Earth.

(a) Show that the star's reflex orbital speed around the center of mass is $\mathbf{v}_\star \approx \mathbf{m_p}\,((2\pi\mathbf{G})/(\mathbf{M}_\star{}^2\mathbf{P}))^{1/3}$.

(b) Compute the Sun's reflex speed $\mathbf{v}_\odot$ (in m/s) caused by Jupiter ($\mathbf{m_J} \approx 9.55 \times 10^{-4}\,\mathbf{M}_\odot$, $\mathbf{P} = 11.86$ yr, $\mathbf{M}_\odot = 1.99 \times 10^{30}$ kg), and find the resulting fractional wavelength shift $\Delta\lambda/\lambda_0$.